TKGR-GPRSCL: Enhance Temporal Knowledge Graph Reasoning with Graph Structure-Aware Path Representation and Supervised Contrastive Learning

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Abstract. Temporal knowledge graph reasoning (TKGR) aims to infer new temporal fact knowledge based on existing ones, which has become a major concern due to its wide demands, especially for time-sensitive modeling scenarios. However, existing temporal knowledge graph representation learning models usually have challenges in capturing complex graph structure-aware information and discriminating different relation expressions for unknown query prediction, which potentially limits the model performance on the TKGR task. In this paper, we propose a novel TKGR model named TKGR-GPRSCL. Specifically, we first introduce a maximum entropy-based random walk sampler, which can sample sufficient paths containing complicated graph structure-aware information in the TKG in form of path embedding. Then, we model the preference of different paths to the target entity, and the correlations between the target entity and its temporal property depending on transformations in the complex plane for encoding temporal fact. Finally, we advance supervised contrastive learning of relation patterns in light of the discrimination of different relation expressions to be preferably leveraged to predict entities for testing the queries, for those unknown queries in particular. We evaluate TKGR-GPRSCL on three TKGR benchmark datasets, and experimental results demonstrate that our model achieves superior performance compared to competitive baselines (All the data and codes are publicly available at https://github.com/shengyp/TKGR-GPRSCL).

Keywords: Temporal knowledge graph reasoning · Graph structure-aware path representation · Supervised contrastive learning

1 Introduction

Temporal knowledge graph reasoning (TKGR) has always been a challenging task in the field of knowledge engineering due to its demand for a high-level comprehension

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of evolving fact knowledge in the knowledge graph. Since several time-sensitive applications, such as semantic search [1], policymaking [5], stock forecasting [6] can be modeled as TKG reasoning diagram, thus resulting in its increasing attention in recent years.

Temporal knowledge graph reasoning (TKGR) aims to predict unknown facts by modeling snapshots of historical KGs, which involves two reasoning settings, i.e., interpolation and extrapolation [20]. More specifically, interpolation methods typically estimate unknown knowledge through the relevant known facts, and extrapolation methods aim to estimate unknown knowledge in the future with relevant historical events. In this paper, we focus on the extrapolation task, i.e., the model needs to answer the query *(Olivia Rodrigo, Release an album, ?, 2023-9-8)* by matching and selecting from all candidate entities based on related historical temporal facts.

Recently, most contemporary researchers have shed light on the path patterns for diverse types of heterogeneous graph representation. One direct motivation behind these studies is to learn each node's embedding by aggregating all involved paths instead of its neighborhoods. Following the line of research work, many efforts have been made to capture temporal evolution information using path-based embedding in TKGs. For example, Kgedl [30] captures the evolution process of TKG by embedding entities, relations, and path structures. Kgedl further applies temporal information to constrain path reasoning and representation learning of entities. RE-NET [10] focuses on encoding one-hop repetitive paths (repetitive facts) in history. However, facing enumerating all paths for each node in the TKG, how to define an appropriate sampler for obtaining and encoding paths for extracting sufficient graph structure-aware information associated with TKG is quite challenging.

Moreover, we notice that complex embedding functions-based TKGR methods typically embed TKGs into complex spaces or special coordinate systems to capture various relational patterns and semantic information. Typically, TComplEx [12] extends the learned representations to the complex space, thus further improving the performance of TKGR. TeRo [31] defines the temporal evolution of entity embedding as a rotation and regards relation as translation. These works motivate us to capture path preference, and temporal information across the target node through transformations in the complex plane.

Several typical relation patterns, exhibit similar (e.g., *Appeal for change in institutions* and *Express intent to institute political reform*), different (e.g., *Appeal for easing of administrative sanctions* and *Appeal for aid*), and opposite (e.g., *Accede to demands for change in leadership* and *Reject request for change in leadership*) relations in the TKGR bechmark dataset, serving to connect different entities play significant roles in the TKG. The focus of CyGNet [35] is on identifying recurring patterns to determine the frequency of similar facts in the history. DE-SimplE [8] incorporates temporal information into diachronic entity embedding for modeling various relation patterns. Along with their achievements, existing research efforts largely focused on how to incorporate diverse relation patterns for TKG representation learning models, and the ignorance of separating the representations of different fact with the patterns, which potentially limit the TKG representation learning for TKGR. More surprisingly, we observe that even supervised contrastive learning models are capacity of discriminating different relation expressions in the perspective of relation patterns.

Based on the aforementioned analysis, in this paper, we proposed a new temporal knowledge graph reasoning model with graph structure-aware path representation and supervised contrastive learning (short-handed as TKGR-GPRSCL) for enhancing TKGR performance. Specifically, TKGR-GPRSCL introduces a maximum entropy-based random walk sampler, which effectively explores the diversity of graph structure-aware information in the TKG. To better encode temporal fact, we make a transformation to the complex plane when model the preference of each path embedding to the target entity, and correlations between the target entity and its temporal feature in form of the score function of each temporal fact in the TKG. Then, we advance supervised contrastive learning of relation patterns to pull together representations belonging to the same class (relation pattern) in embedding space, while simultaneously pushing apart representations of different classes. This allows the discrimination of different relation expressions to be preferably leveraged to predict entities for testing the queries, for those unknown queries in particular. In summary, the main contributions of our work are as follows:

- We propose a novel temporal knowledge graph reasoning model, named TKGR-GPRSCL, which has the capacity to capture complex structure-aware information by encoding paths across entities in the TKG. Meanwhile, our model also obtains correlations between the entity and its temporal property depending on transformations in the complex plane to better model temporal fact representation.
- We propose a novel supervised contrastive learning to discriminate different facts from relation pattern perspectives to improve the model generalization further.
- We conduct extensive experiments on three widely used public TKGR benchmark datasets, which validate the superiority and effectiveness of our method.

2 Prelimanary

2.1 Notations and Task Formulation

Temporal Knowledge Graph (TKG). TKG is a directed graph along with a collection of temporal fact. Let $\mathcal{G}_t = \{\mathcal{E}, \mathcal{R}, \mathcal{T}, \mathcal{F}\}$ be a TKG instance, where $\mathcal{E}$, $\mathcal{R}$, $\mathcal{T}$ represent the set of entities, relations, and timestamps, respectively. $\mathcal{F}$ denotes the set of temporal facts, each of which is represented as a quadruple (s, p, o, t). Here, $s \in \mathcal{E}$ and $o \in \mathcal{E}$ represent the subject and object entities, respectively, and $p \in \mathcal{R}$ denotes the predicate (relation) as a relation type, $t \in \mathcal{T}$ denotes the corresponding timestamp. Under this definition, a TKG can be represented as a knowledge graph (KG) snapshots $\mathcal{G}_t = \{\mathcal{E}, \mathcal{R}, \mathcal{F}_t\}$, where $\mathcal{F}_t$ is the set of temporal facts that occurred at time t.

Semantic Embedding of TKG. For a temporal fact $(s, p, o, t) \in \mathcal{G}_t$, the semantic embedding of subject s and object entity o is defined as $\mathbf{e}_s \in \mathbb{R}^d$ and $\mathbf{e}_o \in \mathbb{R}^d$, respectively. The semantic embedding of relation r is defined as $\mathbf{r}_p \in \mathbb{R}^d$. Meanwhile, the temporal property embedding of subject s, object entity o, and relation r is defined as $\mathrm{t}_s \in \mathbb{R}^d$, $\mathrm{t}_o \in \mathbb{R}^d$, and $\mathrm{t}_r \in \mathbb{R}^d$, respectively.

Adjacency Matrix of Graph. Let a graph $\mathcal{G} = \{\mathcal{E}, \mathcal{R}\}$, where $\mathcal{E}$, $\mathcal{R}$ represent the set of nodes, edges. The adjacency matrix of $\mathcal{G}$ is denoted by $\mathbf{A} \in \{0, 1\}^{|\mathcal{E}|\times|\mathcal{E}|}$. In this paper, we consider TKG $\mathcal{G}_t$'s natural graph features is regard a graph for conducting maximum entropy-based path sampling based on its adjacency matrix $\mathbf{A}$.

Temporal Knowledge Graph Reasoning. Given a TKG instance $\mathcal{G}_t$ with timestamps t range from t_0 to t_n, the objective of temporal knowledge graph reasoning (TKGR) is to predict either the subject in a give query $(?, r, o, t)$ or the object in a given query $(s, r, ?, t)$ with $t > t_n$.

3 Related Work

3.1 Path Patterns and Entropy-Based Sampling Methods in the Graph

In recent years, several models have emerged for path patterns for graph structure learning. GeniePath [22] proposes an adaptive path layer that can guide the breadth and depth exploration of the receptive fields. SPA-GAN [34] conducts the operation of path-based higher-order attention to explore the graph topological information. However, each node in the path influences the target node only through the shortest distance and omits the structure of higher-order neighborhoods. On the other hand, Li *et al.* [16] first proposes the entropy-based sampling for structure information of the graph, which can not only detect the natural or true structure but also measure the complexity of the graph. MinGE [23] considers both feature entropy and structure entropy on graphs, which are carefully designed according to their rich information characteristics. In general, these patterns-based and sampling approaches cannot attend simultaneously to capture the contextual structure of the graph and reduce the memory cost for propagation on larger graphs such as KG. Sun *et al.* [27] introduce a maximal entropy path sampler to alleviate over-expansion issues and preserve the larger graph structure information. Moreover, they prove that sampling an increasing amount of paths close to the infinite paths scenario at an exponential rate from both theoretical and empirical performance aspects. On the benefit of the great potential to capture complex information and computing costs in the graph, we also leverage the above sampling method to discover the diversity of graph structure-aware information in the TKG.

3.2 Temporal Knowledge Graph Representation Learning

There has been a growing number of works on graph neural networks (GNNs) to explore the intrinsic topology relevance between entities or between entities and relations in TKG to obtain high-quality embeddings. RE-NET [10] uses a multi-relational graph aggregator to capture the graph's structural information. RE-GCN [19] employs RGCN to aggregate messages from neighboring entities and utilizes an auto-regressive GRU to model the temporal dependency among temporal facts. However, these models fail to capture the semantic information and correlations between the entity and its temporal property when representing the temporal fact. Recently, TComplEx [12] is proposed to extend the ComplEx and calculate the score of each temporal fact in the complex plane. TeRo [31] borrows ideas from TransE [2] and RotatE [28] models, which defines the temporal evolution of entity embedding as a rotation and regards relation as translation. These works motivate us to consider different path preferences, semantic information, and its temporal property across the target entity in the complex plane and model the score of temporal fact in the TKG.

4 The Proposed Method

As demonstrated in Fig. 1, the overall architecture of the proposed TKGR-GPRSCL framework mainly contains three components: (1) *Maximum Entropy-Based Path Sampler*, which derives paths under the guidance of maximal entropy random walk, which effectively explores the diversity of graph structure-aware information in the TKG. (2) *Temporal Fact Encoding*, which models the influence of path information on the target entity, the correlations across the target entity with its temporal feature through transformations in the complex plane, and trains the categorical cross-entropy loss $\mathcal{L}_{pre}$. (3) *Supervised Contrastive Learning*, which trains the supervised contrastive loss $\mathcal{L}_{SCLL}$ to discriminate fact features from the relation pattern perspective. In the following, we provide a detailed introduction to three components and elaborate on the steps within each component.

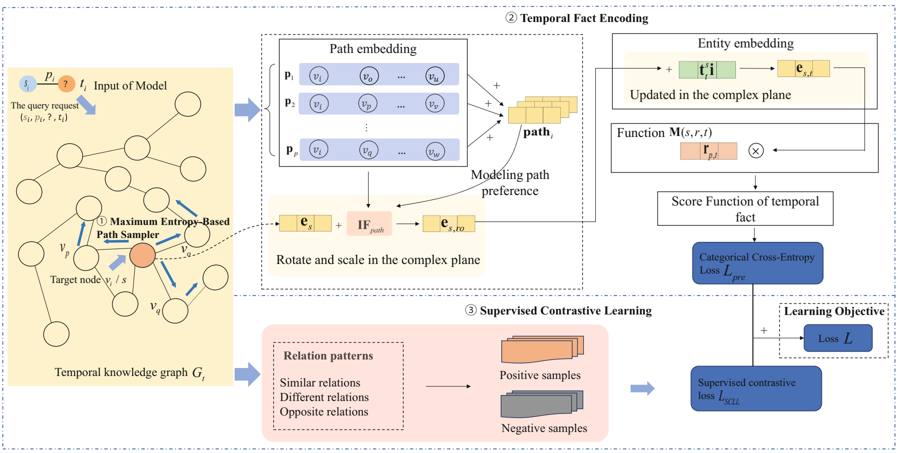


Fig. 1. The overall architecture of the proposed TKGR-GPRSCL model.

4.1 Maximum Entropy-Based Path Sampler

To sample meaningful graph structure-aware paths while a consideration of computing efficiency, inspired by the recent success of entropy-based graph sampling methods [27], we explore the maximum entropy-based random walk sampler, which attempt to collect the paths in the orientation of entropy rate increase in each step and incorporates the eigenvector centrality to estimate the importance of the nodes.

Concretely, consider a path from v_i to v_j with sampling length k that has probability $\mathcal{P}_{i,j}^k = p_{ii_1}p_{i_1i_2}\cdots p_{i_{k-2}i_{k-1}}p_{i_{k-1}j}$, where p_{ij} is an element of the transition matrix. As previously outlined in [4], the maximum entropy rate η of a random walk on a graph can be computed from the transition matrix $\mathcal{P}$ and the stationary distribution π, which is defined as follows:

$$\eta = -\sum_i \pi_i \sum_j p_{ij} ln p_{ij}, \tag{1}$$

Furthermore, the maximum entropy rate of the random walk is bounded by $ln\lambda$, where λ represents the maximum eigenvalue of the adjacency matrix $\mathbf{A}$ corresponding to the TKG $\mathcal{G}$. The sampler builds the transfer probability as $p_{ij} = \frac{A_{i,j} u_j}{\lambda u_i}$ with purpose of maximizing the entropy rate of a walk, where $u = (u_1, u_2, \cdots, u_n)$ is the normalized eigenvector. It can be reformulated in the form of a maximum entropy transition matrix, and it is defined as:

$$\mathbf{P}_u = \frac{\mathbf{D}_u^{-1} \mathbf{A} \mathbf{D}_u}{\lambda}, \tag{2}$$

where $\mathbf{D}_u = \mathrm{diag}(u_1, u_2, \cdots, u_n)$. Under the guidance of the above sampling strategy, the sampler can not only discovers maximum entropy rate but also seeks the sufficient graph structure-aware information associated with TKG $\mathcal{G}$ as a natural graph.

We get the path representation for the target node as follows:

$$\mathbf{path_i} = \frac{1}{p} \sum_{j \in (1,\ldots,p)} \sigma \left(\mathbf{W_{out}} \mathbf{H}_i + \mathbf{b_{out}} \right), \tag{3}$$

where $\mathbf{H}_i = \mathbf{LSTM}(E_{i,j})$, $E_{i,j}$ is the concatenated node embeddings for paths of target node v_i from node v_i to node v_j. The p is the number of paths for target node v_i.

4.2 Temporal Fact Encoding

The semantic and temporal embeddings are considered as a complex vector, where the semantic embedding serves as the real component, while the temporal embedding functions as the imaginary part. Together, they constitute a holistic representation in the complex vector space. At a given timestamp $t \in \{1, 2, \ldots, |\mathcal{T}|\}$, the entities s, o and the relations p can be articulated in the following manner:

$$\mathbf{e}_{s,t} = \mathbf{e}_s + \mathbf{t}_t^s \mathbf{i}, \quad \mathbf{e}_{o,t} = \mathbf{e}_o + \mathbf{t}_t^o \mathbf{i}, \quad \mathbf{r}_{p,t} = \mathbf{r}_p + \mathbf{t}_t^r \mathbf{i}, \tag{4}$$

where the embedding for the subject, object, and relation with current timestamp t are denoted by $\mathbf{e}_{s,t}$, $\mathbf{e}_{o,t}$, and $\mathbf{r}_{p,t}$. The $\mathbf{e}_s$, $\mathbf{e}_0$ and $\mathbf{r}_p$ are denoted for the real part, as well as the $\mathbf{t}_t^s$, $\mathbf{t}_t^o$ and $\mathbf{t}_t^r$ are denoted for the imaginary part. The $\mathbf{i}$ is the base of the imaginary part.

Modeling Path Preference. To determine the path influence $\mathbf{IF}_{path}$ on the entity s, we perform a comprehensive rotation and scaling within the complex plane as follows:

$$\mathbf{IF}_{path} = \mathbf{p}_s + \mathbf{path}_s \otimes \mathbf{p}_s + \mathbf{t}_t^s \otimes \mathbf{p}_s, \tag{5}$$

where $\mathbf{p}_s$ is the path semantic embedding of the entity s. $\mathbf{path}_s$, $\mathbf{t}_t^s$ are the path representation and the temporal embedding of the entity s, respectively. The $\otimes$ represents the complex number multiplication.

Finally, we get the latest entity representation after the comprehensive rotation and scaling within the complex plane as follows:

$$\mathbf{e}_{s,ro} = \mathbf{e}_s + \mathbf{IF}_{path}, \tag{6}$$

Score Function. After the transformations in the complex plane, we update the representation for the entity s in the complex plane as $\mathbf{e}_{s,t} = \mathbf{e}_{s,ro} + \mathbf{t}_t^s\mathbf{i}$. In order to get score function for any quadruple $(s, r, o, t) \in \mathcal{G}$, we compute the complex multiplication of $\mathbf{r}$ and $\mathbf{s}$:

$$\mathbf{M}(s, r, t) = \mathbf{r}_{p,t} \otimes \mathbf{e}_{s,t} = \mathbf{q}_0 + \mathbf{q}_1\mathbf{i}, \tag{7}$$

where $\mathbf{M}(s, r, t)$ represents the rotation and scaling of s under the relation r with timestamp t. The $\mathbf{q}_0$ and the $\mathbf{q}_1$ are the real and imaginary parts of $\mathbf{M}(s, r, t)$, respectively. The score function is as follows:

$$\phi(s, r, o, t) = \mathbf{q}_0\mathbf{e}_s + \mathbf{q}_1\mathbf{path_s}. \tag{8}$$

We define the entity prediction loss function based on Categorical Cross-Entropy Loss as follows:

$$\begin{aligned}\mathcal{L}_{pre} = \sum_{(s,r,o,t)\in\mathcal{G}} \Big[&- \log \left(\exp(\phi(s, r, o, t)) \Big/ \sum_{o'} \exp(\phi(s, r, o', t))\right) \\ &- \log \left(\exp(\phi(o, r^{-1}, s, t)) \Big/ \sum_{s'} \exp(\phi(o, r^{-1}, s', t))\right)\Big],\end{aligned} \tag{9}$$

where o' and s' are the object and subject of the negative quadruples (s, r, o', t) and (o, r^{-1}, s', t) respectively. In the negative quadruples, o$'$ is not o and s' is not s. The r^{-1} is the inverse relation.

4.3 Supervised Contrastive Learning

Inspired by [11], we explore contrastive loss to pull together the representations belonging to the same relation patterns, while simultaneously push apart samples from different relation patterns. For an anchor entity s_i in the temporal fact $f_i(s_i, r_i, o_i, t_i)$, if there exists s_j in a temporal fact $f_j(s_j, r_j, o_j, t_j)$ with more than three neighbor entities that are the same with entity s_i, we refer to relation r_j as a positive sample of relation r_i, otherwise it is considered a negative sample of relation r_i. In a batch dataset $\boldsymbol{I} \equiv \{1 \cdots N\}$, considering a sample with index $i \in \boldsymbol{I}$, the supervised contrastive loss $\mathcal{L}_{SCLL}$ is defined as follows:

$$\mathcal{L}_{SCLL} = \sum_{i\in I} \frac{-1}{|P(i)|} \sum_{p\in P(i)} \log \frac{\exp\left(\boldsymbol{z}_i \cdot \boldsymbol{z}_p/\tau\right)}{\sum_{n\in N(i), n\neq i} \exp\left(\boldsymbol{z}_i \cdot \boldsymbol{z}_n/\tau\right)}, \tag{10}$$

where set $P(i)$ is defined as the indices of all positive samples in the batch dataset $\boldsymbol{I}$ that are distinct from sample with index i. Its size is denoted by $|P(i)|$. $\boldsymbol{z_i} = r_i \oplus t_i^r$ denotes the concatenation of relations representation r_i and temporal embedding t_i^r of relation r_i. $N(i) \equiv \boldsymbol{I} \setminus \{P(i), i\}$, $\cdot$ denotes the inner product. $\tau \in \mathcal{R}^+$ is a scalar temperature parameter.

Table 1. Statistics of datasets in the experiments.

Dataset	#Entity	#Relation	#Triples			#Timestamps	Time Interval
			#Train	#Valid	#Test		
ICEWS14	7,128	230	72,826	8963	8941	365	2014
ICEWS05-15	10,488	251	10	24	240	4017	[2005–2015]
GDELT	500	20	2,735,685	341,961	34,1961	366	[2015–2016]

4.4 Learning Objective

We train our model by jointly minimizing the sum of the aforementioned two losses $\mathcal{L}_{pre}$ and $\mathcal{L}_{SCLL}$:

$$\mathcal{L} = \mathcal{L}_{pre} + \lambda\mathcal{L}_{SCLL} + ||\Theta||_2^2, \tag{11}$$

where λ is a weight that balance the importance of two losses. Θ denotes all trainable model parameters.

5 Experiments and Analysis

5.1 Experimental Settings and Implementation Details

Dataset Description. We evaluate our model and baselines on five real-world TKGs that have been widely used in previous studies, i.e., ICEWS14 [7], ICEWS05-15 [7] and GDELT [15]. ICEWS (Integrated Crisis Early Warning System) and GDELT (Global Database of Events, anguage, and Tone) are event-based TKGs. ICEWS14 and ICEWS05-15 are subsets generated from the Integrated Crisis Early Warning System (ICEWS), which contains a large number of political events with specific timestamps. GDELT is a subset generated from the Global Database of Events and Language Datasets, which contains 20 types of events such as making a public statement, appealing, and consulting. The statistics of these TKGs are shown in Table 1. To ensure a fair comparison, we use the split manner provided by Sun *et al.* [25].

Evaluation Metrics. We adopt evaluation metrics for mean reciprocal rank (MRR) and Hits@k (k = 1, 3, 10) standard metrics for the entity prediction task. MRR is the average reciprocal rank of the correct query answer rank. Hits@k indicates the proportion of correct answers among the top k candidates. Additionally, we reports all experimental results on MRR and Hits@1/3/10 as percentages.

Baselines for Comparison. To comprehensively evaluate our model, we compare our model with two categories of KGR models: (1) *static KG reasoning models*, including DisMult [33], ComplEx [29], Conv-TransE [24], RotatE [13]. (2) *TKGR models*, including TTrans-E [14], TA-DisMult [7], De-SimIE [9], TNTComplEx [12], TITer [26], RE-GCN [19], CEN [18], TiRGN [17], RETIA [21], CENET [32], and logCL [3].

Implementation Details. For the whole of the datasets, the hidden embedding size is set to 600, the learning rate is set to 0.1, the batch size is set to 1000, and the maximum number of epochs is set to 20. For the maximum entropy-based random walk sampler in our model, the number and walk length of the path are set to 1 and 8, respectively. The hidden size of LSTM is set to 160. The hyperparameter λ of the total loss function is set to 0.3. In total, we train our model for 200 epochs and use the Adam optimizer with parameters, with a learning rate of 0.003 and a weight decay of 1e-5. The batch size is set to 256. All of the experiments are processed on a Linux server with the GeForce GTX 3090 GPU.

5.2 Performance Comparison

Table 2 shows the performance comparison results between our TKGR-GPRSCL model and different types of baselines for entity prediction task on three TKGR datasets. The best results are highlighted in bold and the second-best results are indicated underlined format. We summarize our findings as:

Table 2. Experimental results of entity prediction on the ICEWS14, ICEWS05-15, and GDELT datasets. The best and second best results in each column is boldfaced and underlined, respectively. Improvements over the best baseline are shown in the last row.

Models	ICEWS14				ICEWS05-15				GDELT			
	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
TransE (2014)	15.44	10.91	17.24	23.92	17.95	13.12	20.71	29.32	8.68	5.58	9.96	17.13
ComplEx (2016)	32.54	23.43	36.13	50.73	32.63	24.01	37.50	52.81	16.96	11.25	19.52	32.35
Conv-TransE (2019)	33.80	25.40	38.54	53.99	33.03	24.15	38.07	54.32	16.20	10.85	18.38	30.86
RotatE (2019)	21.31	10.26	24.35	44.75	24.71	13.22	29.04	48.16	13.45	6.95	14.09	25.99
TTransE (2016)	13.72	2.98	17.70	35.74	15.57	4.80	19.24	38.29	5.50	0.47	4.94	15.25
TA-DisMult (2018)	25.80	16.94	29.74	42.99	24.31	14.58	27.92	44.21	12.00	5.76	12.94	23.54
DE-SimIE (2020)	33.36	24.85	37.15	49.82	35.02	25.91	38.99	52.75	19.70	12.22	21.39	33.70
NTNComplEx (2020)	34.05	25.08	38.50	50.92	27.54	9.52	30.80	42.86	19.53	12.41	20.75	33.42
TITer (2021)	40.87	32.28	45.45	57.10	47.69	37.95	52.92	65.81	15.46	10.98	15.61	24.31
RE-GCN (2021)	40.39	30.66	44.96	59.21	48.03	37.33	53.85	68.27	19.64	12.42	20.90	33.69
CEN (2022)	42.20	32.08	47.46	61.31	46.84	36.38	52.45	67.01	20.39	12.96	21.77	34.97
TiRGN (2022)	44.04	33.83	48.95	63.84	50.04	39.25	56.13	70.71	21.67	13.63	23.27	37.60
RETIA (2023)	42.76	32.28	47.77	62.75	47.26	36.64	52.90	67.76	20.12	12.76	21.45	34.49
CENET (2023)	39.02	29.62	43.23	57.49	41.95	32.17	46.93	60.43	20.23	12.69	21.70	34.92
LogCL (2023)	48.87	37.76	54.71	70.26	57.04	46.07	63.72	77.87	23.75	14.64	25.60	42.33
TKGR-GPRSCL (ours)	**51.13**	**41.60**	**60.72**	**74.86**	**58.65**	**46.89**	**66.79**	**79.08**	**24.22**	**15.01**	**26.32**	**43.29**
Improvement (%)	4.62	10.17	10.98	6.54	2.82	1.77	4.81	1.55	1.97	2.52	2.81	2.26

- The results demonstrate that our model outperforms all baselines on ICEWS14, ICEWS05-15 and GDELT datasets across all evaluation metrics. On the three ICEWS datasets, SKG embedding methods demonstrate inferior performance compared to TKGR-GPRSCL, due to their inability to capture temporal dynamics.
- The entity prediction effect of TKGR-GPRSCL on the ICEWS14 and ICEWS05-15 datasets has been enhanced, which substantiates the important role of TKGR-GPRSCL in modeling the influence of path information on target nodes. Therefore, it is essential to quantify the influence of nodes through entropy rate on paths, and the way in which we use supervised contrastive learning to model similar, different

or opposite relationships between different relations in the TKG facilitates entity prediction. With regard to the GDELT dataset, the performance of TKGR-GPRSCL has also improved. This performance enhancement can be attributed to the vast data volume of the GDELT dataset, which contains a significant amount of data but comparatively fewer relations. However, traditional baseline models often fail to adequately capture the relationships between these limited relations. In contrast, TKGR-GPRSCL approach succeeds in modeling these relationships, thus resulting in improved the performance on the GDELT dataset.

5.3 Model Ablation and Effectiveness Analysis

We further explore how different components in TKGR-GPRSCL contribute to the entity prediction performance of TKGR-GPRSCL, we conduct an ablation study. Overall, the ablation experiments suggest that each component has a significant positive impact on TKGR-GPRSCL.

Ignore Path Information Influence (w/o Path). Without considering the influence of path information on target entity in the complex plane, the overall effect will be reduced by 3.66%, 4.55%, 4.05%, and 1.51%. TKGR-GPRSCL evaluates the influence of each entity in the path information on the target node based on the entropy rate, so as to obtain the path information of the target node. TKGR-GPRSCL ignores the influence of unimportant node information on the path, and retains the influence of important node information, so as to obtain the path information. Through w/o Path, we can see that considering the influence of path information on the target entity is an indispensable part of TKGR-GPRSCL, which significantly enhances the ability of our model to predict entities.

Ignore Supervised Contrastive Learning (w/o SCL). Without considering the influence of supervised contrastive learning, the overall effect will be reduced by 2.72%, 3.23%, 4.05%, and 1.37%. Relations are an important part of TKG, and we use supervised contrastive learning to model the similar, different or opposite relationships between different relations in the TKG. Through w/o SCL, we can see that adopting supervised contrastive learning to model the relationships of relations play an important part role in TKGR-GPRSCL.

Ignore Path Information Influence and Supervised Contrastive Learning (w/o Path and SCL). When the path information and supervised contrastive learning parts

Table 3. Ablation study of TKGR-GPRSCL model on the ICEWS14 dataset. “w/o” means “without”.

	MRR	Hits@1	Hits@3	Hits@10
TKGR-GPRSCL w/o-Path	49.47	37.05	56.67	73.35
TKGR-GPRSCL w/o-SCL	50.41	38.37	57.84	73.49
TKGR-GPRSCL w/o-Path and SCL	49.23	36.87	57.02	72.56
TKGR-GPRSCL (ours)	53.13	41.60	60.72	74.86

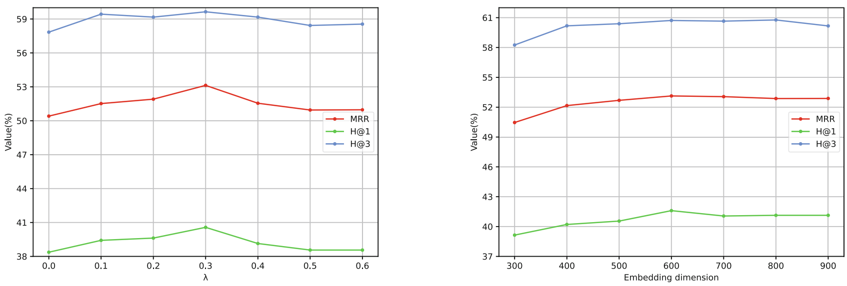
of the model are not considered, the experimental results are generally degraded. Therefore, the path information and supervised contrastive learning parts play an active and important role in TKGR-GPRSCL.

5.4 Studies on Hyperparameter Sensitivity

We further explore how different components conducted experiments on the following hyperparameters to assess the hyperparameter sensitivity of TKGR-GPRSCL.

The Weights λ. We select λ from the set SPSVERBc10.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6SPSVERBc2 and conduct experiments on the ICEWS14 dataset. Figure 2 (a) shows the experimental performance with different λ. With the increase of λ from 0.0 to 0.3, the performance of the model has been continuously improved. But when λ is greater than 0.3, the performance shows a downward trend. Considering the performance under the different values of λ, we choose the value of 0.3 for λ, because when λ is 0.3, the performance of all experimental evaluation metrics reaches the best.

The Embedding Dimension. The embedding dimension selected from the set SPSVERBc1300, 400, 500, 600, 700, 800, 900SPSVERBc2 were used to perform experiments on the ICEWS14 dataset. Figure 2 (b) shows the experimental performance of different embedding dimensions. We can see that with the increase of the dimension from 300 to 600, the performance gradually improves. When the dimension is greater than 600, the performance tends to be flat. Therefore, we chose the embedding dimension of 600, because it strikes a balance between achieving the best performance and minimizing time and space costs.



(a) Weight λ (b) Embedding dimension

Fig. 2. The influence of different hyperparameters.

6 Conclusion

In this paper, we propose a novel temporal knowledge graph reasoning model, named TKGR-GPRSCL. With the introduced maximum entropy-based random walk sampler, TKGR-GPRSCL is capable of sampling sufficient and effective paths reflecting complex graph structure-aware information in the TKG. Meanwhile, TKGR-GPRSCL also achieves correlations between the entity and its temporal property depending on transformations in the complex plane to better model temporal fact representation. Supervised contrastive learning is proposed to discriminate different facts from relation pattern perspectives for improving the generalization performance for testing queries, for those atypical relation classes in particular. Experiment results on three widely used benchmark datasets on the TKGR task demonstrate the effectiveness of TKGR-GPRSCL compared to competitive baselines. For future work, instead of the LSTM network, a more effective path aggregation network to generate path embeddings is worth exploring.

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